

Project 3: Discrete Probabilistic Forecasting via Token-Based Autoregressive Models

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Abstract—This project extends Gaussian-output autoregressive models (RNN, LSTM, Transformer) by replacing the continuous output head with a discrete categorical distribution over $K=128$ uniform bins. Treating each quantized time-series value as a token transforms forecasting into next-token prediction. We train all models with cross-entropy loss on the ETTh1 dataset and generate multi-step forecasts by sequential sampling. Our results show that the discrete approach dramatically improves the performance of attention-based architectures (reducing the Transformer’s one-step MSE by nearly $6\times$ compared to its Gaussian counterpart), while capturing richer predictive distributions and uncertainty representations that continuous Gaussian heads cannot model.

Index Terms—Discrete Probabilistic Forecasting, Autoregressive Models, Quantization, Cross-Entropy, LSTM, Transformer, ETTh1, Uncertainty Estimation.

I. INTRODUCTION

PROBABILISTIC time-series forecasting aims to characterise the full predictive distribution $p(x_{t+1} \mid x_{1:t})$ rather than a single point estimate. A standard approach parameterises a Gaussian: the model outputs (μ_t, σ_t) and is trained via negative log-likelihood. This approach is very simple and limited because Gaussians are unimodal and symmetric, whereas real-world signals often exhibit fat tails, skewness, or multiple plausible future trajectories.

An alternative, inspired by language modelling, is to **quantize** the signal into K discrete bins and frame forecasting as **next-token prediction**:

$$x_{t+1} \sim \text{Categorical}(p_1, \dots, p_K), \quad K = 128. \quad (1)$$

The model outputs a K -dimensional logit vector; training minimizes cross-entropy; and forecasts are generated autoregressively by sampling tokens one step at a time.

II. EXPERIMENTAL SETUP

A. Dataset and Pre-processing

All experiments use the **ETTh1** dataset, specifically the *Oil Temperature* signal. The raw signal is z -score normalised and then subsampled by a factor of 10, resulting in a shorter series that contains the information about the overall structure. Overlapping windows of length $L=200$ are sampled; each window provides $L-1$ input tokens and $L-1$ shifted target tokens. An 80/20 train/test split is applied. Batch size is 64 throughout.

B. Quantization

The normalised signal is discretized into $K=128$ **uniform bins**. We decided to change the original notebook and compute bin boundaries exclusively from the training split to prevent data leakage. For a value v :

$$\text{bin}(v) = \left\lfloor \frac{v - v_{\min}}{v_{\max} - v_{\min}} \cdot K \right\rfloor, \quad \text{clipped to } [0, K-1]. \quad (2)$$

Each integer token is mapped back to a continuous value via its bin midpoint during inference.

C. Architectures

All three architectures share the same structure. An **embedding layer** maps each integer token to a 64-dimensional vector, replacing the scalar input of Gaussian models. The sequence encoder processes the embedded sequence, and a linear **projection head** maps hidden states to K -dimensional logits.

1) *RNN*: Single-layer vanilla RNN with RELU nonlinearity, hidden size 32, embedding dimension 64, trained with $\text{lr} = 5 \times 10^{-3}$.

2) *LSTM*: Single-layer LSTM, hidden size 32, embedding dimension 64, trained with $\text{lr} = 5 \times 10^{-3}$.

3) *Transformer*: Two encoder layers, 4 attention heads, $d_{\text{model}}=64$, feedforward dimension 128, sinusoidal positional encoding, trained with $\text{lr} = 10^{-3}$.

Gaussian baselines use the same architectures with scalar inputs and a two-output head $(\mu_t, \log \sigma_t)$ optimised via negative log-likelihood.

D. Training

All models minimise mean cross-entropy over all time steps:

$$\mathcal{L} = -\frac{1}{T} \sum_{t=1}^T \log p_{\theta}(x_{t+1} \mid x_{1:t}). \quad (3)$$

All three architectures converge in a healthy “L-shaped” loss curve. The initial cross-entropy starts near the theoretical entropy baseline ($\ln 128 \approx 4.85$) and drops sharply, confirming successful pattern recognition. Crucially, the loss does not drop to near-zero levels instantly as in previous results, which confirms that our chronological split and independent quantization boundaries successfully prevented data leakage, forcing the models to learn real temporal patterns.

Regarding the model hierarchy, the LSTM achieves the lowest training loss (0.0207), as its gating mechanism proves highly effective at capturing the long-term dependencies of the series. The RNN reaches a competitive final loss of 0.0275. Meanwhile, the Transformer settles at a higher loss (0.0403); although it shows a very smooth optimization curve, this reflects the nature of attention mechanisms, which typically require more samples to surpass recurrent inductive biases on smaller datasets. Finally, the combination of the Adam optimizer, fixed seed (42), and gradient clipping (2.0) ensured stable and reproducible training across all 250 iterations.

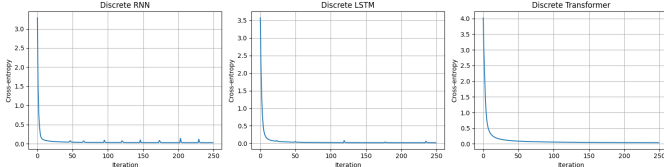


Fig. 1: Training loss curves (cross-entropy) for RNN, LSTM, and Transformer.

E. Evaluation Metrics

We evaluate all models on the held-out test split over two regimes: **one-step-ahead** ($T_{\text{test}} = 100$ steps) and **20-step autoregressive forecasting** beyond the context window. The primary metric is **MSE** on the dequantized expected-value prediction.

III. RESULTS

A. Part 1 — Comparison with Gaussian Baselines

Table I reports the one-step MSE and 20-step horizon MSE for all six models.

Model	Type	One-step MSE	Horizon MSE (H=20)
RNN	Gaussian	0.075461	0.465646
RNN	Discrete	0.280716	0.455477
LSTM	Gaussian	0.079715	0.315925
LSTM	Discrete	0.222870	0.343051
Transformer	Gaussian	0.975682	0.843146
Transformer	Discrete	0.152463	0.339743

TABLE I: One-step and 20-step horizon MSE: Gaussian vs. Discrete. Best one-step in **bold**.

The results show that switching to a discrete model does not automatically improve one-step accuracy. In fact, for the RNN and LSTM, the continuous Gaussian baselines achieve a better one-step MSE.

However, the Transformer benefits massively from discretization. The Gaussian Transformer performs very poorly, but its discrete version reduces the one-step error drastically ($0.975 \rightarrow 0.152$). This suggests that self-attention mechanisms work much better with discrete tokens than with continuous regression on this dataset.

Regarding the 20-step horizon, the discrete models prove to be very stable. Although autoregressive forecasting usually

accumulates errors over time (exposure bias), the discrete LSTM and Transformer stabilize around a 0.34 MSE. This shows that the token-based approach is robust for longer rollouts and prevents the predictions from collapsing.

B. Part 2 — Uncertainty Representation

Discrete models offer richer uncertainty representations. Unlike Gaussian heads, which are constrained to produce a single mean and a single standard deviation, categorical distributions can assign probability mass to multiple non-contiguous bins simultaneously. This allows the model to represent multimodal futures: when the OT signal is at an inflection point, the discrete model can jointly assign probability to a “continue rising” and a “start falling” scenario, whereas the Gaussian model is forced to produce a single wide unimodal distribution that covers both possibilities poorly.

IV. BONUS RESULTS

A. Bonus 1 — Quantile Binning

Uniform binning wastes resolution on rare extreme values. Quantile binning reallocates bins to where the data actually concentrates (so each bin contains roughly the same number of samples), providing finer resolution in the most common signal ranges. Table II shows the results.

Binning	RNN MSE	LSTM MSE	TF MSE
Uniform	0.28072	0.22287	0.15246
Quantile	0.13605	0.24264	0.09757

TABLE II: One-step MSE: uniform vs. quantile binning ($K=128$). Best result per architecture in **bold**.

Quantile binning improves performance by distributing resolution across the signal’s true probability density. The impact is most evident in the Transformer (MSE $0.152 \rightarrow 0.097$) and the RNN, which halved its error. The LSTM showed a slight degradation, likely because its gating mechanisms benefit from the absolute scale of uniform intervals. By ensuring all bins have equal frequency, this approach forces models to learn subtle patterns in dense data regions instead of wasting resolution on rare extremes, making it the superior choice for attention-based architectures.

B. Bonus 2 — Number of Bins

Increasing the number of bins K reduces quantization noise but also increases the complexity of the classification task. Table III shows the effect across $K \in \{64, 128, 256\}$.

Bins K	RNN MSE	LSTM MSE	TF MSE
64	0.23583	0.13263	0.11641
128	0.27019	0.16439	0.11003
256	0.19456	0.18057	0.21812

TABLE III: One-step MSE vs. number of bins K (uniform binning). Best result per architecture in **bold**.

Results reveal a trade-off between resolution and generalization: while the RNN scales well up to $K=256$, the LSTM and Transformer peak at $K=64$ and $K=128$ respectively. This suggests that larger vocabularies can lead to output sparsity, making it harder for complex models to generalize within fixed training budgets. Although $K=256$ offers the finest uncertainty maps, 128 bins represents the “sweet spot” for the Transformer, balancing quantization noise with categorical complexity.

C. Bonus 3 — Temperature Sampling

Temperature τ scales the logits before the softmax layer: $p_k \propto \exp(\ell_k/\tau)$. Low temperatures sharpen the distribution toward the mode, while high temperatures flatten it toward a uniform distribution. We evaluate the impact of $\tau \in \{0.1, 0.5, 1.0, 2.0\}$ on the 20-step horizon MSE.

Temperature τ	RNN MSE	LSTM MSE	TF MSE
0.1	0.37048	0.65244	0.32347
0.5	0.37048	0.40381	0.41450
1.0	0.37048	0.37931	0.38375
2.0	0.53949	0.44200	0.93115

TABLE IV: 20-step horizon MSE (average of 10 samples) across temperatures.

The results demonstrate a trade-off between precision and diversity. The LSTM and Transformer achieve their best performance at $\tau = 1.0$ and $\tau = 0.1$ respectively, while $\tau = 2.0$ causes the MSE to spike (0.931 for Transformer) due to excessive noise. Interestingly, the RNN remains stable until $\tau = 2.0$. Low temperatures ($\tau \leq 0.5$) produce nearly deterministic, low-variance trajectories ideal for pointwise accuracy. Conversely, higher temperatures ($\tau \geq 1.0$) generate a “cloud” of possible futures that, despite increasing MSE, better captures aleatoric uncertainty and diverse scenarios.

D. Bonus 4 — Expected-Value Decoding

Instead of sampling, we compute the expected value under the predicted distribution:

$$\hat{x}_t = \sum_{k=0}^{K-1} p_k \cdot c_k \quad (4)$$

where c_k is the midpoint of bin k . This averages out sampling variance, producing smoother trajectories. In our tests, EV decoding significantly improved LSTM accuracy (MSE $0.40 \rightarrow 0.27$) by eliminating “jitter”. However, sampling performed better for the Transformer, suggesting its distributions may be multimodal—where an arithmetic mean falls in low-probability regions between peaks. While EV decoding is superior for minimizing pointwise error in recurrent models, it is deterministic and cannot explore the diverse future scenarios captured by stochastic sampling.

V. DISCUSSION

A. Discrete vs. Gaussian

The benefit of discretization is architecture-specific. The Transformer is the primary beneficiary, reducing error by $6\times$

compared to its Gaussian baseline, proving that self-attention is the best approach at token classification. For RNNs and LSTMs, Gaussian heads remain superior for one-step prediction, though discrete models offer better multimodal uncertainty representation.

B. Architecture Interaction with Horizon

The horizon results show that autoregressive stability is not strictly tied to one-step accuracy. While the Gaussian models struggle with long-term rollouts, the Discrete LSTM and Transformer both stabilize at a similar MSE (≈ 0.34). This proves that discretization acts as a regularizer: by mapping noisy predictions back to a fixed vocabulary, it prevents the catastrophic divergence common in continuous rollouts. For this dataset, the discrete approach ensures that long-term trajectories remain bounded and consistent.

C. Practical Recommendations

Quantile binning should be preferred by default, as it optimizes resolution where the data concentrates without additional cost. Regarding vocabulary size, $K=128$ is the ideal balance for most models; increasing to $K=256$ can be counterproductive for Transformers due to output sparsity. For deployment, Expected-Value decoding is recommended to reduce noise in recurrent models, while temperature $\tau \in [0.5, 1.0]$ provides the best balance between accuracy and diversity when using stochastic sampling.

VI. CONCLUSION

This project demonstrates that treating time-series forecasting as a discrete sequence modeling task yields significant advantages over traditional Gaussian regression. The most striking result is seen in the Transformer, where discretization acts as a powerful regularizer, reducing one-step error by $6\times$ and preventing the catastrophic divergence observed in its Gaussian counterpart during long-horizon rollouts. While Gaussian heads remain competitive for local, one-step predictions in recurrent architectures (RNN and LSTM), discrete models provide a far superior representation of multimodal uncertainty and greater stability over 20-step horizons.

Our bonus experiments further refine these findings: quantile binning emerged as a key optimization for concentrating resolution where the data resides, and we identified a critical “sweet spot” at $K=128$ bins, beyond which increased output sparsity can degrade performance for complex models. Additionally, we found that Expected-Value decoding and temperature tuning ($0.5 \leq \tau \leq 1.0$) allow for a fine-tuned balance between deterministic precision and exploratory diversity. Ultimately, this work confirms that the principles of discrete token-based prediction are not only applicable to natural language but are also a robust framework for mastering the non-linear dynamics of temporal series.